

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Physics

2. **Subject Name:** General Properties of Matter Lab
Credits: 2

L-T-P: 0-0-3

3. **Pre-requisites:** None

4. **Syllabus and reference books:**

Syllabus:

1. Determination of the co-efficient of viscosity of water by its streamline flow through a capillary tube.
2. To study the flow of water through a capillary tube as a function of pressure head using two tubes of the same length but different radii.
3. Determination of the surface tension using Capillary Rise tube of the liquids at room temperature.
4. Determination of the Young's modulus of the different metallic bars by the flexure method.
5. Modulus of rigidity by torsional pendulum
6. Momentum of Inertia
7. Velocity of sound in metals
8. To determine the figure of merit of a given dead-beat type galvanometer by lamp and scale arrangement.
9. Study of the normal and anomalous Hall effect
10. Velocity of electromagnetic waves by using luminous diode and photodiode (light receiver).

Reference Books:

Lab manual